Assignment 2 Answers

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Part A

Blackjack is a card game played between at least 2 participants. For this example, there will only be two participants. The first (and simpler) participant is the dealer. The dealer doesn't has zero choices in the game, and is therefore not very interesting. The second participant in the game is the player. The player actively makes decisions. The dealer gives two card to himself (one face up and one face down). The dealer gives two card to the player, both of which are face up. The player has two choices: to receive another card from the dealer (hit), or to not receive another card from the dealer (stand). The goal of the player is to hold cards who's sum is closer to 21 than the dealer's sum of cards.

Considering all previous conditions. The following can be said.

- Environment's goal: to acquire a sum that is closer to 21 than the dealer's sum.
- State/observation space: the player's sum and the dealer's face up card.
- Action space: the choice to hit or stand.
- Transition dynamics: the dealer gives another card to the player.
- Termination conditions
  - If the player acquires a natural (without hitting) sum of 21. A reward of +1.5 is given.
  - The dealer's sum is greater than 21. A reward of +1 is given.
  - The player's sum is greater than 21. A reward of -1 is given.

- After the player stands, their sum is greater than the dealer's sum. A reward of +1 is given.
- After the player stands, the dealer's sum is greater than the player's sum. A reward of -1 is given.
- After the player stands, their sum is equal to the dealer's sum (push). A reward of +0 is given.

Part B

I have created the blackjack CLI agent to implement the Q-learning algorithm. The following code snippit is found withtin the training function.

```
if terminated or truncated:
    target = reward
else:
    target = reward + gamma * np.max(Q[next_state])

Q[state + (action,)] += alpha * (
    target - Q[state + (action,)]
)
```

Please run my BlackjackCLI.py script to train the algorithm yourself.

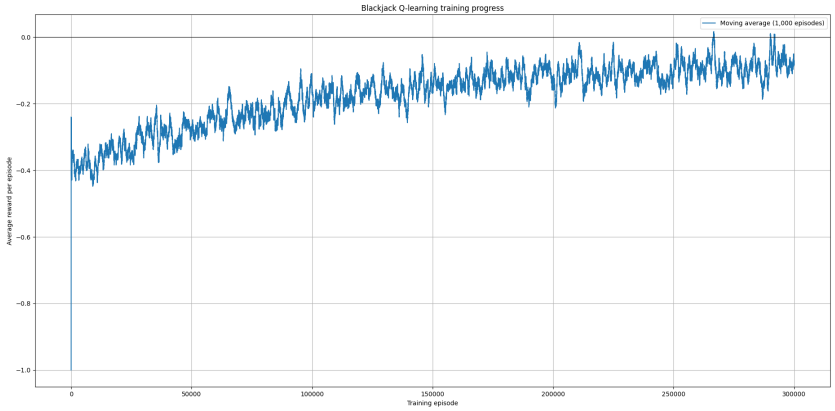
Part C

My baseline parameters for AI training are:

- alpha = 0.1
- gamma = 0.95
- episilon = 1.0
- $\text{episilon}_{\text{min}} = 0.05$
- $\text{episilon}_{\text{decay}} = 0.99999$
- episodes = 300,000

The resulting graph shows how the algorithm preformed. After quickly jumping to an average reward of -0.4 from the first few rounds, the algorithm slowly climbs to an average reward of 0. There is a plateau from episodes

200,000 to 300,000. Perhaps additional training from episode 200,000 is unnecessary?



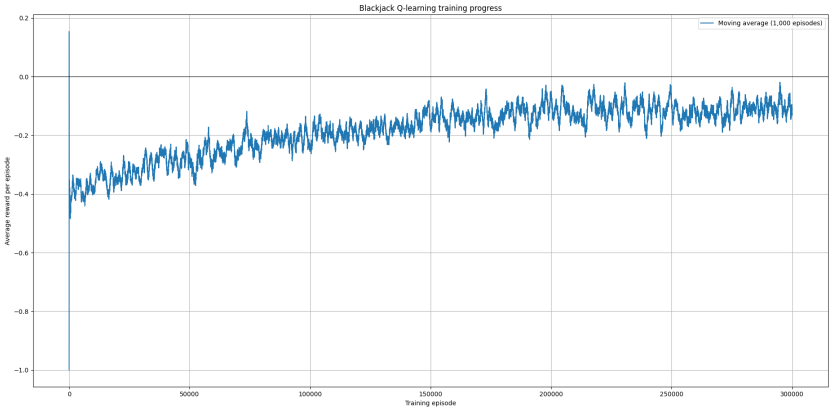
Part D

All other parameters are the same as in part C except.

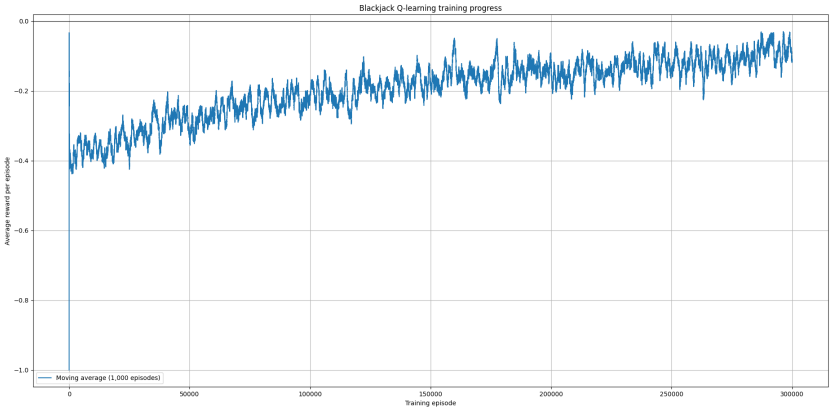
- In the first graph, alpha = 0.2
- In the second graph, alpha = 0.5
- In the third graph, alpha = 0.9

Generally speaking, it seems increasing the alpha parameters makes the variance larger. In other words, the graph seems to be more violtile. This is reasonable conclusion because the alpha parameter represents how valued new experience is compared to the sum of all previous experience.

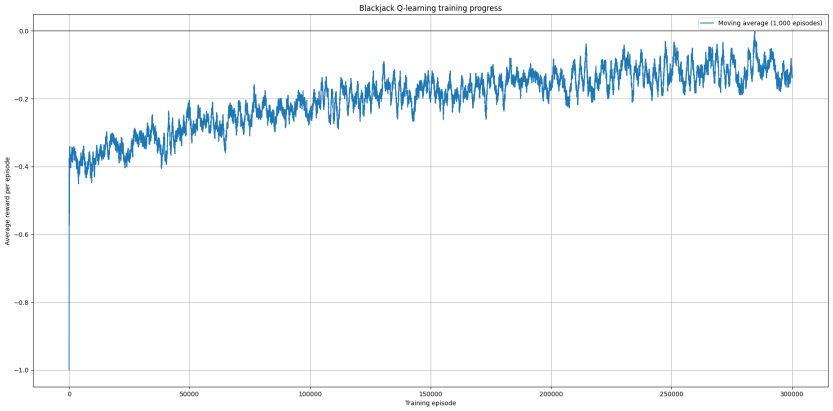
First Graph



Second Graph



Third Graph

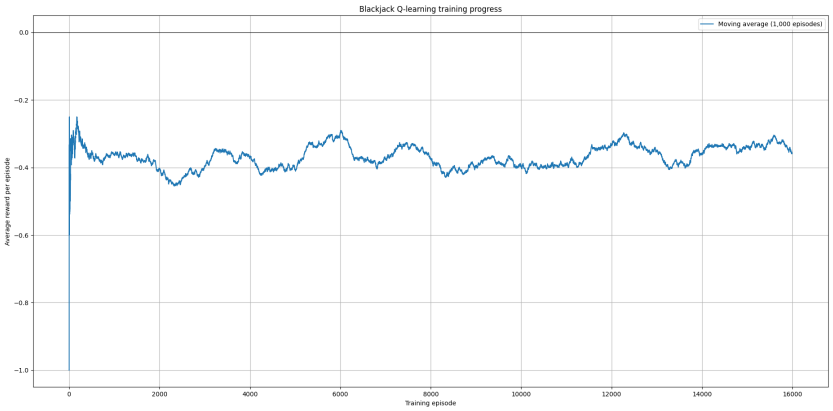


Part E

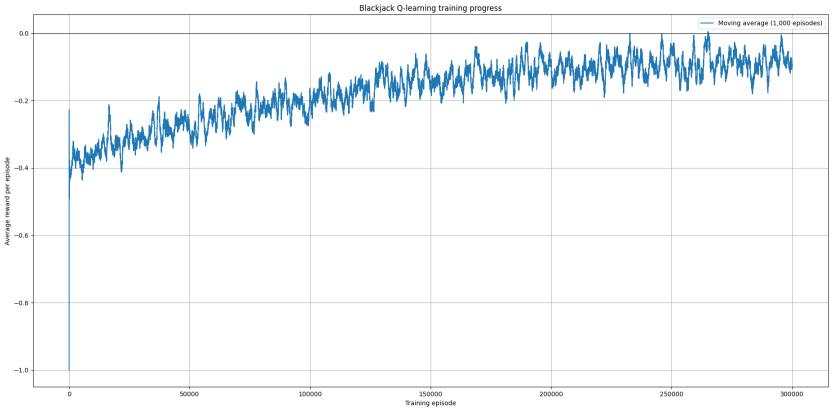
All other parameters are the same as in part C except.

- In the first graph, gamma = 0.5
- In the second graph, gamma = 0.9
- In the third graph, gamma = 0.99

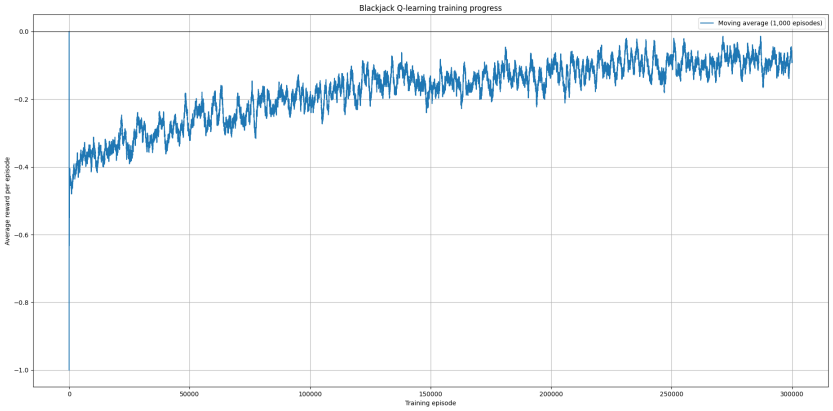
First Graph



Second Graph



Third Graph



Part F

Now, I will alter the episilon decay. All other parameters are the same as in part C except.

- In the first graph, episilon decay = 0.9999
- In the second graph, episilon decay = 0.999
- In the third graph, episilon decay = 0.99

First Graph

Second Graph

Third Graph